UC SAN DIEGO / ECE 251C

7

Multiresolution analysis and wavelet filtering

Executive summary

Multiresolution analysis organizes a signal into nested approximation spaces and mutually orthogonal detail spaces. We formulate its defining properties, show how the detail spaces recover the full signal space, and express the resulting decomposition in scaling and wavelet coefficients. The two-scale equations then connect this geometry to a discrete algorithm. For Haar, explicit overlap integrals give sums and differences of neighboring coefficients. Reversing the masks turns these formulas into convolution, and downsampling places the outputs on the coarse grid. Repeating the same procedure on the approximation branch produces the multilevel discrete wavelet transform.

7.1 From local averages to a discrete algorithm

The Haar construction in Lecture 6 separated each pair of neighboring fine cells into an average and a difference. We now ask which properties of that construction allow the same idea to work with other scaling functions. The answer is the structure of the approximation spaces and the way their bases behave under dilation.

The distinction between functions and their coefficients is central. The spaces V_i and W_i describe continuous-domain approximation and detail functions. A digital implementation stores the coefficient sequences of these functions. The goal is to obtain coarse and detail coefficients directly from fine coefficients, without reacquiring the signal or evaluating the continuous-domain basis functions.

Recall the geometric decomposition. If $P_i f$ is the orthogonal projection of f onto V_i, then

$$V_{i+1} = V_i \oplus W_i, \qquad P_{i+1} f = P_i f + (P_{i+1} f - P_i f).$$

The difference belongs to W_i and is orthogonal to V_i. Thus the finer approximation contains both the old information and one new increment.

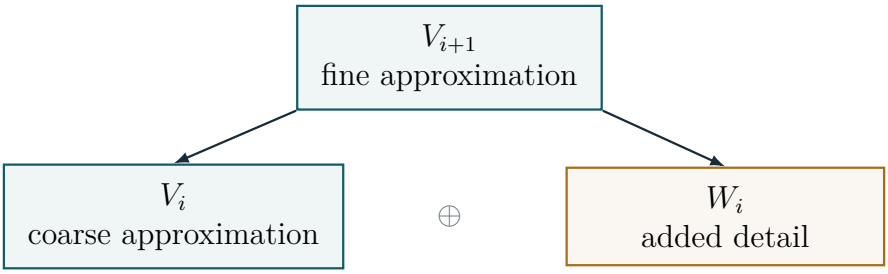


Figure 7.1: One finer representation, split into two orthogonal components. The two-scale expansions will turn these projections into filtering operations.

7.2 What defines a multiresolution analysis?

The essential properties of the Haar construction are the nesting of its approximation spaces, their behavior under dilation, and the existence of an orthonormal set of translates. These properties do not require box-shaped functions. Retaining them as axioms allows us to separate the multiresolution structure from the particular choice of scaling function.

The multiresolution framework was introduced by Mallat and Meyer around 1988. It gives a common description of the approximation spaces, wavelet bases, and filtering operations that appeared in the Haar example.

Definition. An *orthonormal multiresolution analysis* (MRA) is a nested sequence of closed subspaces

$$\cdots \subset V_{-1} \subset V_0 \subset V_1 \subset \cdots \subset L^2(\mathbb{R})$$

with the following properties:

1. **Fine resolutions are complete:**

$$\overline{\bigcup_{i\in\mathbb{Z}} V_i}^{L^2} = L^2(\mathbb{R}).$$

   Any finite-energy signal can be approximated arbitrarily well at sufficiently fine resolution.

2. **No information persists at every coarse scale:**

$$\bigcap_{i\in\mathbb{Z}} V_i = \{0\}.$$

   No nonzero finite-energy signal belongs to all the spaces.

3. **Changing scale changes resolution:**

$$f \in V_0 \quad \Longleftrightarrow \quad f(2^i \cdot) \in V_i.$$

Compressing the coordinate by a factor of two moves up one level.

4. **Integer translates stay in the reference space:**
$$f \in V_0 \quad \Longrightarrow \quad f(\cdot - n) \in V_0, \qquad n \in \mathbb{Z}.$$

5. **One scaling function supplies the coordinates.** There is a function $\phi \in V_0$ such that $\{\phi(\cdot - n)\}_{n\in\mathbb{Z}}$ is an orthonormal basis of V_0.

A subspace is *closed* when it contains the L^2 limit of every convergent sequence of its elements. This is the condition that guarantees the existence of an orthogonal projection onto the subspace. By contrast, the closure of the union in the first axiom expresses an approximation property: a signal need not belong to any one V_i, provided elements of progressively finer spaces approximate it arbitrarily well.

For Haar, $\phi = \mathbf{1}_{[0,1)}$ and V_i consists of finite-energy functions constant on cells of width 2^{-i}. Other constructions use nonorthogonal stable bases (Riesz bases) or redundant stable families (frames). We keep the orthonormal setting here.

7.3 Detail spaces are orthogonal increments

The information added between two successive approximation spaces should be independent of the information already retained. We express this independence by orthogonality. The detail space is therefore the orthogonal complement of V_i *inside* V_{i+1}, rather than the set-theoretic difference of these spaces:

$$W_i := \{q \in V_{i+1} : \langle q, v\rangle_{L^2} = 0 \text{ for every } v \in V_i\}. \tag{7.1}$$

Equivalently, $W_i = V_{i+1} \cap V_i^{\perp}$. Orthogonality here uses

$$\langle f, g\rangle_{L^2} = \int_{\mathbb{R}} f(t)g^*(t)\,\mathrm{d}t, \qquad \|f\|_{L^2}^2 = \langle f, f\rangle_{L^2}.$$

As before, the *second* function is conjugated.

Why does this give a direct sum? For $v \in V_{i+1}$, write

$$v = P_i v + (v - P_i v).$$

The first term belongs to V_i. The second belongs to V_{i+1} and is orthogonal to V_i, so it belongs to W_i. A function in both V_i and W_i is orthogonal to itself and hence has zero norm. Thus the decomposition is unique:

$$\boxed{V_{i+1} = V_i \oplus W_i.} \tag{7.2}$$

The same projection argument as in Lecture 5 gives

$$\|v\|_{L^2}^2 = \|P_i v\|_{L^2}^2 + \|v - P_i v\|_{L^2}^2.$$

Thus the squared norms of the two components add to the squared norm of the original function. No cross term remains because the components belong to orthogonal subspaces. This is the function-space form of the energy identity for an orthogonal change of coordinates.

Different detail levels are orthogonal. If $k < i$, then

$$W_k \subset V_{k+1} \subset V_i, \qquad W_i \perp V_i.$$

Therefore $W_k \perp W_i$. A fine detail is orthogonal not only to its own coarse approximation, but also to every earlier detail already contained in that approximation.



Figure 7.2: Approximation spaces are nested. Detail spaces are the mutually orthogonal increments between them; the detail spaces are not themselves nested.

7.4 Recovering the approximation spaces from the details

Repeatedly substitute $V_{k+1} = V_k \oplus W_k$. For integers $I < i$,

$$V_i = V_I \oplus W_I \oplus W_{I+1} \oplus \cdots \oplus W_{i-1}. \tag{7.3}$$

For example, $V_3 = V_0 \oplus W_0 \oplus W_1 \oplus W_2$. The upper detail index is 2, not 3: W_3 would supply the additional information needed to reach V_4.

The starting level I is arbitrary. Letting it move toward coarser resolutions suggests that all the earlier detail spaces should recover V_i:

$$\boxed{V_i = \overline{\bigoplus_{k=-\infty}^{i-1} W_k}^{L^2}}. \tag{7.4}$$

Here the direct sum consists of finite sums of elements from the mutually orthogonal spaces W_k; the closure includes their L^2 limits. The following argument explains why no component is missing.

Proof. Let U_i denote the right-hand side. Each W_k, $k < i$, lies in V_i, so $U_i \subset V_i$. For $v \in V_i$, let $r = v - \mathrm{proj}_{U_i} v$. Then $r \in V_i$ and $r \perp W_k$ for every $k < i$.

For any $I < i$, decompose r using equation (7.3). Orthogonality to $W_I, \ldots, W_{i-1}$ forces all its detail components to vanish. Hence $r \in V_I$ for every $I < i$. By nesting it also belongs to the spaces with $I \geq i$. The MRA intersection condition gives $r = 0$, so $v \in U_i$. □

The argument uses ordinary orthogonal projection onto a closed subspace: the residual is perpendicular to that subspace. The intersection condition rules out a residual shared by every coarse space. Thus the approximation spaces determine the detail spaces through orthogonal complements, and the detail spaces recover the approximation spaces through equation (7.4).

Since the union of the V_i is dense, equation (7.4) also gives

$$L^2(\mathbb{R}) = \overline{\bigoplus_{i\in\mathbb{Z}} W_i}^{L^2}. \tag{7.5}$$

Alternatively, stop at the fixed reference resolution 0 and keep its approximation:

$$\boxed{L^2(\mathbb{R}) = V_0 \oplus \overline{\bigoplus_{i=0}^{\infty} W_i}^{L^2}}. \tag{7.6}$$

The space V_0 contains the information from all negative-resolution detail spaces. Keeping a scaling-function expansion in V_0 replaces those infinitely many coarse detail levels by one approximation component. It does not discard that component of the signal.

7.5 From subspaces to wavelet synthesis

The scaling function gives an orthonormal basis at every resolution:

$$\phi_{i,n}(t) \coloneqq 2^{i/2}\phi(2^i t - n), \qquad n \in \mathbb{Z}.$$

To describe the detail spaces, choose a wavelet ψ such that $\{\psi(\cdot - n)\}_{n\in\mathbb{Z}}$ is an orthonormal basis of W_0, and set

$$\psi_{i,n}(t) \coloneqq 2^{i/2}\psi(2^i t - n).$$

The Haar wavelet from Lecture 6 has this property. For a general construction, merely choosing $\psi \in W_0$ is not enough: its translates must span W_0 and be orthonormal.

Why does the same normalization work at every scale? For any two functions u, v, the substitution $s = 2^i t$ gives

$$\int_{\mathbb{R}} 2^{i/2}u(2^i t)\, 2^{i/2}v^*(2^i t)\, \mathrm{d}t = 2^i 2^{-i} \int_{\mathbb{R}} u(s)v^*(s)\, \mathrm{d}s.$$

Thus normalized dilation preserves inner products. It maps V_0 onto V_i and V_1 onto V_{i+1}, so it also maps their orthogonal complements W_0 and W_i onto one another. The dilated families remain orthonormal bases.

Coefficients and reconstructed components. Define

$$a_i[n] \coloneqq \langle f, \phi_{i,n} \rangle_{L^2}, \qquad d_i[n] \coloneqq \langle f, \psi_{i,n} \rangle_{L^2}. \tag{7.7}$$

As in Lecture 6, $d_i[n]$ denotes a detail coefficient, while $q_i(t)$ denotes the reconstructed detail function. With this distinction, the projections take the form:

$$P_i f = \sum_{n \in \mathbb{Z}} a_i[n] \phi_{i,n}, \qquad q_i = \operatorname{proj}_{W_i} f = \sum_{n \in \mathbb{Z}} d_i[n] \psi_{i,n}.$$

Square brackets denote coefficients; t is the continuous coordinate.

The mutually orthogonal spaces in equation (7.6) show that

$$\{\phi_{0,n} : n \in \mathbb{Z}\} \ \cup \ \{\psi_{i,n} : i \geq 0, \ n \in \mathbb{Z}\}$$

is an orthonormal basis of $L^2(\mathbb{R})$. The full signal is

$$\boxed{f(t) = \sum_{n \in \mathbb{Z}} a_0[n] \phi_{0,n}(t) + \sum_{i=0}^{\infty} \sum_{n \in \mathbb{Z}} d_i[n] \psi_{i,n}(t).} \tag{7.8}$$

The series converges in L^2. It describes *synthesis*: each coefficient weights a known basis function, and the resulting approximation and detail components are added. *Analysis* computes these coefficients from the signal. The first detail level is $i = 0$, since W_0 is the first increment beyond V_0.

7.6 The two-scale coefficients are inner products

The inclusion $V_0 \subset V_1$ has a concrete consequence: the scaling function itself can be expanded in the finer basis. Likewise, $\psi \in W_0 \subset V_1$. Orthogonal projection therefore supplies both expansions:

$$\phi(t) = \sum_{n \in \mathbb{Z}} g_0[n] \phi_{1,n}(t), \qquad g_0[n] = \langle \phi, \phi_{1,n} \rangle_{L^2}, \tag{7.9}$$

$$\psi(t) = \sum_{n \in \mathbb{Z}} g_1[n] \phi_{1,n}(t), \qquad g_1[n] = \langle \psi, \phi_{1,n} \rangle_{L^2}. \tag{7.10}$$

The factor $\sqrt{2}$ is already part of $\phi_{1,n}(t) = \sqrt{2}\phi(2t - n)$. The sequences g_0, g_1 are coordinates in this normalized basis, not the heights of the functions in the plots.

Calculate the Haar coefficients by projection. Take

$$\phi = \mathbf{1}_{[0,1)}, \qquad \psi = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}.$$

Only the two finer boxes inside $[0,1)$ can overlap either function. Hence

$$g_0[0] = \sqrt{2}\int_0^{1/2} 1\,\mathrm{d}t = \frac{1}{\sqrt{2}}, \qquad g_0[1] = \sqrt{2}\int_{1/2}^{1} 1\,\mathrm{d}t = \frac{1}{\sqrt{2}},$$
$$g_1[0] = \sqrt{2}\int_0^{1/2} 1\,\mathrm{d}t = \frac{1}{\sqrt{2}}, \qquad g_1[1] = \sqrt{2}\int_{1/2}^{1} (-1)\,\mathrm{d}t = -\frac{1}{\sqrt{2}}.$$

All other coefficients vanish. Thus

$$g_0[n] = \frac{\delta[n] + \delta[n-1]}{\sqrt{2}}, \qquad g_1[n] = \frac{\delta[n] - \delta[n-1]}{\sqrt{2}}. \tag{7.11}$$

The projection calculation recovers exactly the masks found by inspection in Lecture 6. It also gives a definition that does not rely on the shape of a box: for any orthonormal MRA and associated wavelet, the two-scale coefficients are inner products with the finer scaling basis.

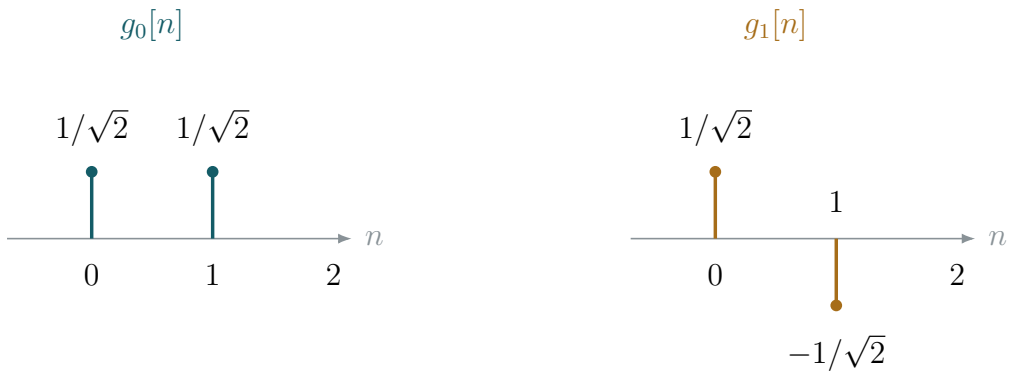


Figure 7.3: The same finer boxes build either an average or a difference. The minus sign in $g_1[1]$ comes from the negative half of the wavelet.

7.7 What information is available at the finest resolution?

Suppose our acquisition system has supplied the Haar coefficients at resolution 3:

$$a_3[n] = \langle f, \phi_{3,n}\rangle_{L^2}, \qquad f_3 = P_3 f = \sum_{n\in\mathbb{Z}} a_3[n]\phi_{3,n}.$$

These coefficients determine f_3, not necessarily f. The difference $f - f_3$ is information outside the acquired resolution. Our analysis will reorganize the information in f_3 without introducing additional measurements.

Coarser coefficients are already determined. Because $\phi_{2,m}$ and $\psi_{2,m}$ belong to V_3, both are orthogonal to $f - f_3$. Consequently

$$\langle f, \phi_{2,m}\rangle_{L^2} = \langle f_3, \phi_{2,m}\rangle_{L^2}, \qquad \langle f, \psi_{2,m}\rangle_{L^2} = \langle f_3, \psi_{2,m}\rangle_{L^2}.$$

Thus the coefficients $a_2[m]$ and $d_2[m]$ are already determined by the acquired approximation f_3. The part of f discarded in forming f_3 is orthogonal to all the basis functions used at these coarser resolutions.

Expand the fine representation inside the first inner product:

$$\begin{aligned} a_2[m] &= \left\langle \sum_{n\in\mathbb{Z}} a_3[n]\phi_{3,n}, \phi_{2,m} \right\rangle_{L^2} \\ &= \sum_{n\in\mathbb{Z}} a_3[n]\langle \phi_{3,n}, \phi_{2,m}\rangle_{L^2}. \end{aligned} \tag{7.12}$$

The coefficient $a_3[n]$ is not conjugated: it appears in the *first* argument. In the same way,

$$d_2[m] = \sum_{n\in\mathbb{Z}} a_3[n]\langle \phi_{3,n}, \psi_{2,m}\rangle_{L^2}. \tag{7.13}$$

Each output coefficient is therefore a weighted sum of the input coefficients, with weights given by overlaps between basis functions at neighboring resolutions. For Haar, only two overlaps are nonzero. The same calculation is valid for infinite expansions by continuity of the inner product; the finite partial sums converge in L^2.

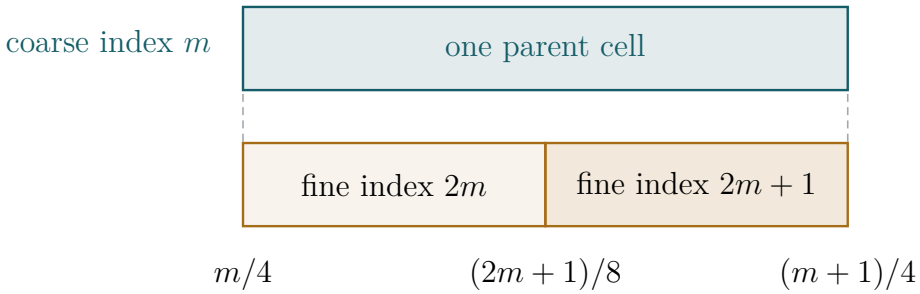


Figure 7.4: At resolutions 2 and 3, one parent interval contains two children. The factor $2m$ comes from expressing the parent location in the finer grid.

7.8 The low-pass weights: an overlap calculation

To calculate the low-pass weight in equation (7.12), we compare the supports of a fine box and a coarse box. The Haar functions are real, so conjugation has no effect in this calculation:

$$\langle \phi_{3,n}, \phi_{2,m}\rangle_{L^2} = 2^{5/2}\int_{\mathbb{R}} \phi(8t-n)\phi(4t-m)\,\mathrm{d}t.$$

First put $u = 8t$, so $\mathrm{d}t = \mathrm{d}u/8$. Then put $v = u - 2m$, so $\mathrm{d}u = \mathrm{d}v$. The limits remain $-\infty$ and $+\infty$ in both changes of variable:

$$\begin{aligned}\langle \phi_{3,n}, \phi_{2,m} \rangle_{L^2} &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi(u-n)\phi(u/2-m)\,\mathrm{d}u \\ &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi\big(v-(n-2m)\big)\phi(v/2)\,\mathrm{d}v. \end{aligned} \tag{7.14}$$

The factor is $2^{5/2}/8 = 2^{-1/2}$. All dependence on the indices is now through the integer $r = n - 2m$.

The first box is one on $[r, r+1)$; the second is one on $[0, 2)$. Their product integrates to the length of their overlap. That length is one for $r = 0, 1$ and zero for every other integer r.

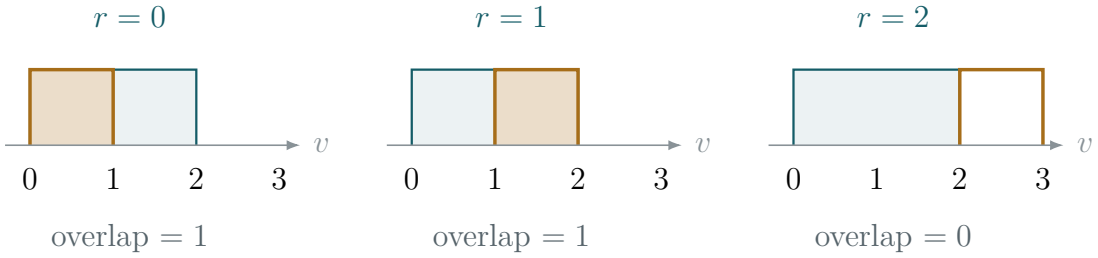


Figure 7.5: The broad box is $\phi(v/2)$; the narrow box is $\phi(v-r)$. A contact at one endpoint contributes zero to the integral.

Thus the overlap is exactly a shifted filter tap:

$$\langle \phi_{3,n}, \phi_{2,m} \rangle_{L^2} = \begin{cases} 1/\sqrt{2}, & n = 2m \text{ or } 2m+1, \\ 0, & \text{otherwise} \end{cases} = g_0[n-2m]. \tag{7.15}$$

Substitution into equation (7.12) yields

$$a_2[m] = \sum_n a_3[n] g_0[n-2m] = \frac{a_3[2m] + a_3[2m+1]}{\sqrt{2}}.$$

The familiar sum of two neighboring coefficients has therefore been derived from the continuous-domain projection. Its weights and its pairing of indices are determined by the overlaps of a parent basis function with its two children.

7.9 The high-pass weights: signed overlaps

The detail coefficient uses a wavelet in place of the coarse box. Begin with the full integral:

$$\langle \phi_{3,n}, \psi_{2,m} \rangle_{L^2} = 2^{5/2} \int_{\mathbb{R}} \phi(8t-n)\psi(4t-m)\,\mathrm{d}t.$$

With $u = 8t$ and then $v = u - 2m$, the calculation is

$$\begin{aligned}\langle \phi_{3,n}, \psi_{2,m} \rangle_{L^2} &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi(u-n)\psi(u/2-m)\,\mathrm{d}u \\ &= \frac{1}{\sqrt{2}} \int_{\mathbb{R}} \phi\big(v-(n-2m)\big)\psi(v/2)\,\mathrm{d}v. \end{aligned} \tag{7.16}$$

The prefactor is unchanged. The difference is that the broad function now has a sign:

$$\psi(v/2) = \begin{cases} 1, & 0 \le v < 1, \\ -1, & 1 \le v < 2, \\ 0, & \text{otherwise.} \end{cases}$$

For $r = n - 2m = 0$, the narrow box covers its positive half, giving integral 1. For $r = 1$, it covers the negative half, giving -1. All other integer translates have zero overlap.



Figure 7.6: The highlighted region is the product $\phi(v-r)\psi(v/2)$. Its signed area distinguishes the left child from the right child.

Consequently,

$$\langle \phi_{3,n}, \psi_{2,m} \rangle_{L^2} = \begin{cases} 1/\sqrt{2}, & n = 2m, \\ -1/\sqrt{2}, & n = 2m+1, \\ 0, & \text{otherwise} \end{cases} = g_1[n-2m]. \tag{7.17}$$

Putting these weights into equation (7.13) gives

$$d_2[m] = \sum_n a_3[n] g_1[n-2m] = \frac{a_3[2m] - a_3[2m+1]}{\sqrt{2}}.$$

The high-pass output measures the contrast between the two children. Equal child coefficients give zero detail. For real signals, the sign distinguishes the direction of the contrast: it is positive when the left coefficient is larger and negative when the right coefficient is larger.

7.10 Why convolution reverses the filters

Our projection formulas contain $g_k[n-2m]$. Standard convolution instead uses a tap indexed by *output location minus input location*:

$$(x * h)[\ell] = \sum_{n\in\mathbb{Z}} x[n]h[\ell - n].$$

The order of subtraction matters. To put the projection formulas into this form, define the real Haar analysis filters

$$\begin{aligned} h_0[k] &:= g_0[-k] = \frac{\delta[k] + \delta[k+1]}{\sqrt{2}}, \\ h_1[k] &:= g_1[-k] = \frac{\delta[k] - \delta[k+1]}{\sqrt{2}}. \end{aligned} \tag{7.18}$$

Then $h_k[2m-n] = g_k[n-2m]$, so

$$a_2[m] = \sum_n a_3[n]g_0[n-2m] = \sum_n a_3[n]h_0[2m-n] = (a_3 * h_0)[2m], \tag{7.19}$$

$$d_2[m] = \sum_n a_3[n]g_1[n-2m] = \sum_n a_3[n]h_1[2m-n] = (a_3 * h_1)[2m]. \tag{7.20}$$

Thus the analysis filters are the time-reversed two-scale masks. This reversal is required by the convolution convention; it is not a change in the projection being computed.

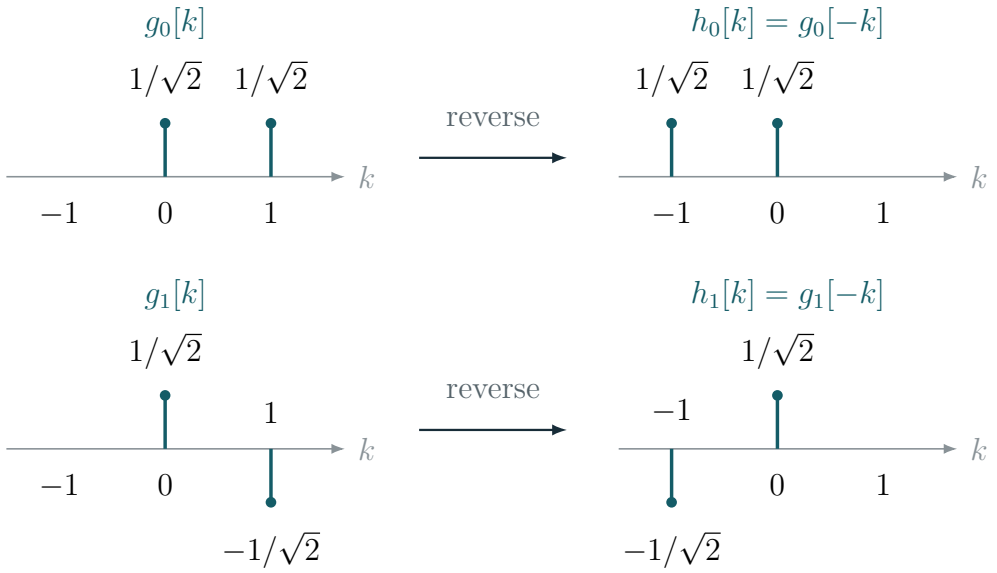


Figure 7.7: Reversal moves the tap at 1 to -1 without changing its value. In particular, the negative high-pass tap moves to the left of the origin.

With this alignment, the filters use the entries $a_3[\ell]$ and $a_3[\ell+1]$. A negative tap index is therefore an indexing convention for the stored array, not a request for an

additional measurement. In a causal streaming implementation, an appropriate delay would be included instead.

7.11 Filtering followed by downsampling

Filtering produces a value at every fine-grid index. The projection needs only the values at the even indices $2m$, because the coarse grid has half as many locations per unit interval.

Definition. Downsampling a sequence z by two gives

$$((\downarrow 2)z)[m] := z[2m].$$

This retains the even-indexed entries and labels them by the coarse index m. It does not average neighboring entries. The averaging or differencing has already been done by the filter.

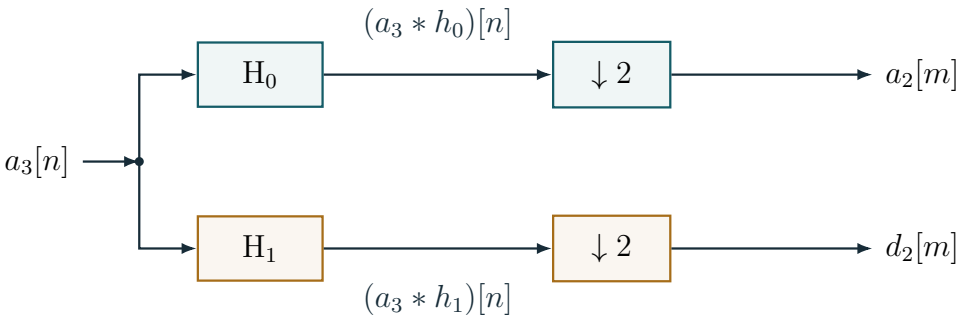


Figure 7.8: The two-channel analysis bank. The top branch computes approximation coefficients; the bottom computes detail coefficients. Each filter block is followed by its own downsampler.

For the Haar filters, write the two operations separately:

$$(a_3 * h_0)[n] = \frac{a_3[n] + a_3[n+1]}{\sqrt{2}}, \qquad (a_3 * h_0)[2m] = \frac{a_3[2m] + a_3[2m+1]}{\sqrt{2}},$$
$$(a_3 * h_1)[n] = \frac{a_3[n] - a_3[n+1]}{\sqrt{2}}, \qquad (a_3 * h_1)[2m] = \frac{a_3[2m] - a_3[2m+1]}{\sqrt{2}}.$$

The positions $2m, 2m+1$ are exactly the two children identified by the overlap calculations.

What has downsampling discarded? One branch alone is insufficient: a sum does not determine two numbers. The sum *and* difference do. Adding and subtracting the outputs gives

$$a_3[2m] = \frac{a_2[m] + d_2[m]}{\sqrt{2}}, \qquad a_3[2m+1] = \frac{a_2[m] - d_2[m]}{\sqrt{2}}.$$

Thus keeping both branches loses no information about f_3. For a finite array of even length, the two half-length outputs together contain the original number of coefficients.

7.12 The same filters work at every resolution

The calculations at resolutions 3 and 2 illustrate a scale-independent rule. The two-scale equations have the same coefficients after a common dilation and translation, so one pair of filters can be used at every stage. We now derive this statement explicitly.

Substitute $2^i t - m$ for t in equation (7.9) and multiply by $2^{i/2}$:

$$\begin{aligned}\phi_{i,m}(t) &= 2^{i/2} \sum_{k\in\mathbb{Z}} g_0[k]\sqrt{2}\,\phi\big(2(2^i t - m) - k\big) \\ &= \sum_{k\in\mathbb{Z}} g_0[k]\, 2^{(i+1)/2}\phi\big(2^{i+1}t - (2m+k)\big) \\ &= \sum_{k\in\mathbb{Z}} g_0[k]\phi_{i+1,2m+k}(t) = \sum_{n\in\mathbb{Z}} g_0[n-2m]\phi_{i+1,n}(t).\end{aligned}$$

The last equality is the index change $n = 2m + k$. Repeating the same dilation in equation (7.10) gives

$$\psi_{i,m} = \sum_{n\in\mathbb{Z}} g_1[n-2m]\phi_{i+1,n}.$$

Keep track of conjugation. Take the inner product of f with these expansions. A coefficient in the second argument is conjugated:

$$\begin{aligned}a_i[m] &= \sum_n g_0^*[n-2m]\, a_{i+1}[n], \\ d_i[m] &= \sum_n g_1^*[n-2m]\, a_{i+1}[n].\end{aligned} \tag{7.21}$$

For real Haar masks this is precisely the calculation above. If the two-scale masks are complex, the correct analysis filters include conjugation as well as reversal:

$$h_0[k] = g_0^*[-k], \qquad h_1[k] = g_1^*[-k]. \tag{7.22}$$

Therefore the general projection rule is

$$\boxed{a_i[m] = (a_{i+1} * h_0)[2m], \qquad d_i[m] = (a_{i+1} * h_1)[2m].} \tag{7.23}$$

The filters do not change with i. Changing level changes the meaning of one coefficient-grid step: it corresponds to a different distance in the continuous domain.

For an orthonormal scaling and wavelet construction, these formulas compute exact projections. Choosing two arbitrary filters does not, by itself, guarantee such a construction. Here the filters inherit their properties from the bases whose overlaps defined them.

7.13 A complete Haar analysis by hand

Continue the signal from Lecture 6 with values $(1, 3, 5, 7, 7, 7, 5, 5)$ on successive eighths of $[0, 1)$ and zero elsewhere. Because it lies in V_3, its acquisition is exact. A normalized basis function on an eighth-cell has height $\sqrt{8}$, so a cell value c gives coefficient $c/\sqrt{8}$:

$$a_3 = \frac{1}{\sqrt{8}}(1, 3, 5, 7, 7, 7, 5, 5).$$

Apply the pair formulas at each level. The first two pairs give

$$a_2[0] = \frac{1+3}{\sqrt{8}\sqrt{2}} = 1, \qquad d_2[0] = \frac{1-3}{\sqrt{8}\sqrt{2}} = -\frac{1}{2},$$
$$a_2[1] = \frac{5+7}{\sqrt{8}\sqrt{2}} = 3, \qquad d_2[1] = \frac{5-7}{\sqrt{8}\sqrt{2}} = -\frac{1}{2}.$$

The equal pairs $(7, 7)$ and $(5, 5)$ have zero details. Continuing yields

resolution i	a_i	d_i
2	$(1, 3, 7/2, 5/2)$	$(-1/2, -1/2, 0, 0)$
1	$(2\sqrt{2}, 3\sqrt{2})$	$(-\sqrt{2}, 1/\sqrt{2})$
0	(5)	(-1)

For example, $a_1[0] = (1+3)/\sqrt{2} = 2\sqrt{2}$ and $d_1[0] = (1-3)/\sqrt{2} = -\sqrt{2}$. At the final split, $a_0[0] = (2\sqrt{2} + 3\sqrt{2})/\sqrt{2} = 5$ and $d_0[0] = -1$.

These are *coefficients*, not cell heights. Substituting them into the synthesis formula gives

$$f = 5\phi - \psi - \sqrt{2}\psi_{1,0} + \frac{1}{\sqrt{2}}\psi_{1,1} - \frac{1}{2}\psi_{2,0} - \frac{1}{2}\psi_{2,1}. \tag{7.24}$$

These coefficients are exact inner products, not estimates obtained by fitting the plotted values. Because the signal belongs to V_3, the filtering computation and the continuous-domain expansion contain exactly the same information.

The first split, shown explicitly. Column m pairs fine indices $2m, 2m+1$ with the outputs $a_2[m], d_2[m]$.

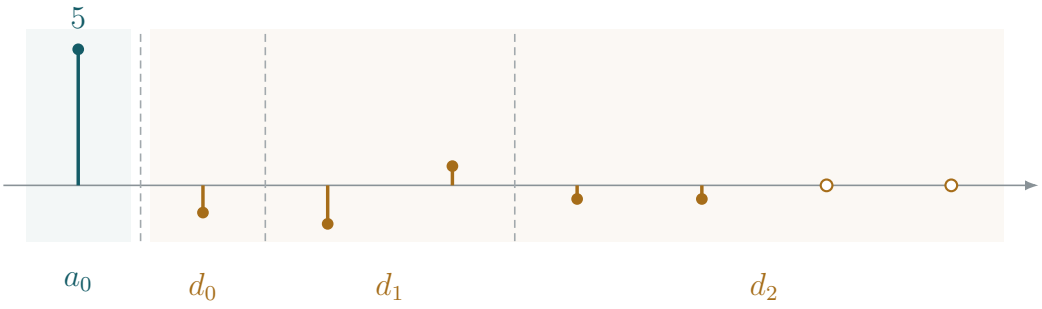


Figure 7.9: The eight coefficients from the table, grouped by resolution. Open circles denote zero detail coefficients, not discarded entries.

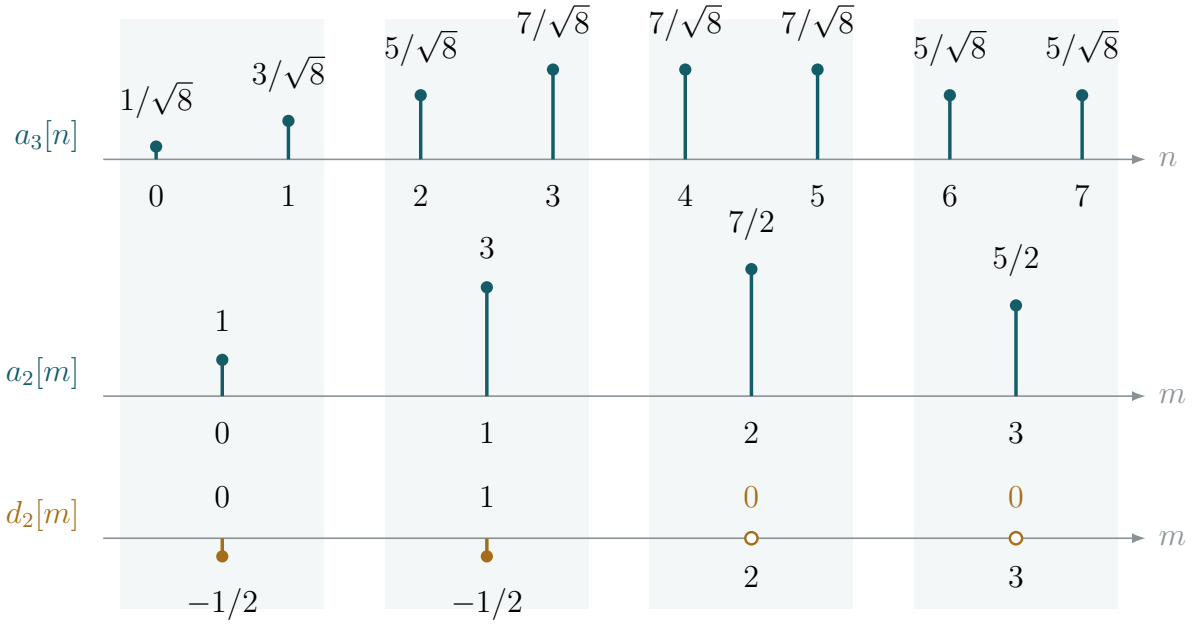


Figure 7.10: One input pair gives its sum and difference divided by $\sqrt{2}$. The rows share vertical units; open circles are zero detail coefficients.

Coefficients are not function heights. The scaling functions have height 2, so a_2 gives quarter-cell values $(2, 6, 7, 5)$. A detail coefficient $-1/2$ multiplies a wavelet of height $+2, -2$, restoring the values $-1, +1$:

$$q_2 = -\tfrac{1}{2}\psi_{2,0} - \tfrac{1}{2}\psi_{2,1}, \qquad f_3 = f_2 + q_2.$$

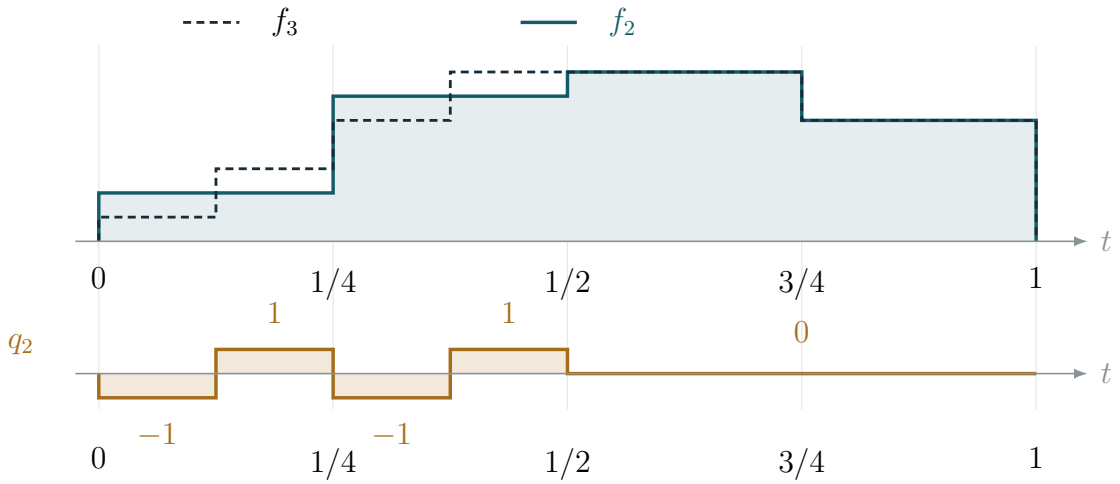


Figure 7.11: The same step in the signal domain. Above: f_3 (dashed) and f_2 (solid), whose quarter-cell heights are $2a_2 = (2, 6, 7, 5)$. Below: $q_2 = f_3 - f_2$, with eighth-cell values $(-1, 1, -1, 1, 0, 0, 0, 0)$. Both plots use the same vertical units.

7.14 Iterating only the approximation branch

Once a_3 has been split into a_2 and d_2, store d_2 and apply the same analysis bank to a_2. This produces a_1, d_1. One further split produces a_0, d_0.

The standard DWT retains d_2 without decomposing it again. This sequence already describes the increment between V_2 and V_3. The next stage separates the information still contained in a_2 into a coarser approximation and its detail. Repeating this procedure produces the multilevel organization of the coefficients.

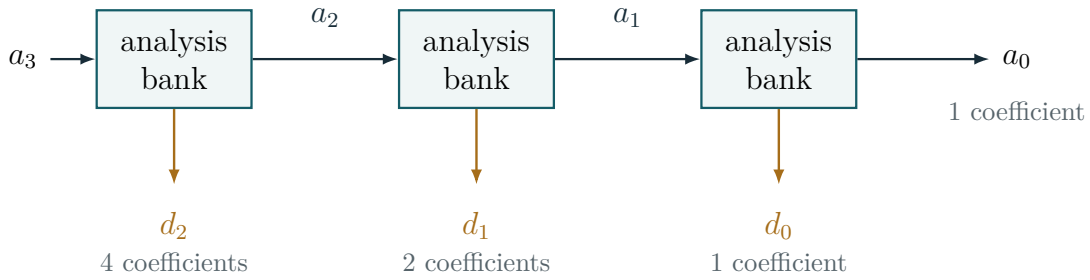


Figure 7.12: Three stages for an eight-coefficient input. The retained outputs have lengths $4, 2, 1, 1$, so their total remains eight. The arrows show computations on coefficient sequences, not new measurements of f.

More generally, start with a finest available sequence a_I and stop at a coarser level $I_0 < I$. The output is

$$a_{I_0}, \quad d_{I_0}, d_{I_0+1}, \ldots, d_{I-1}.$$

These sequences give the expansion

$$\boxed{P_I f = \sum_n a_{I_0}[n]\phi_{I_0,n} + \sum_{i=I_0}^{I-1}\sum_n d_i[n]\psi_{i,n}.} \tag{7.25}$$

Only levels $i < I$ are determined by a_I. To obtain d_I, we would need the additional information at resolution $I + 1$.

This cascade is the analysis half of a perfect-reconstruction filter bank: the recursive analysis procedure of the discrete wavelet transform (DWT).

The connection

Two-scale expansions become operations on coefficients. Convolution requires reversed analysis filters; the coarse grid requires downsampling. Keeping both branches preserves the acquired approximation, and repeating only the coarse branch organizes that information by resolution.